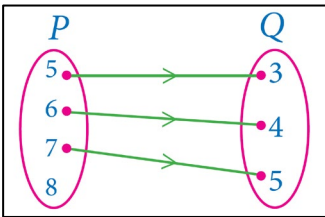
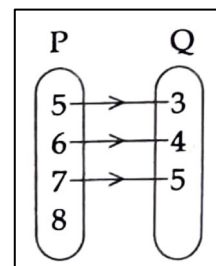


10TH MATHS PUBLIC EXAM QUESTIONS COLLECTIONS**2 MARK QUESTIONS EM NEW (2026 - 2027)****CHAPTER -1.RELATIONS AND FUNCTIONS**

- If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B. (Sep. 2019 & Aug. 2022)
 - Show that the function $f: N \rightarrow N$ defined by $f(m) = m^2 + m + 3$ is one-one function. (Sep.2019)
 - If $A = \{1,3,5\}$ and $B = \{2,3\}$ then find. Show that $n(A \times B) = n(B \times A) = n(A) \times n(B)$ (Sep. 2021)
 - Let $A = \{1, 2,3,4, \dots 45\}$ and R Be the relation defined as “is square of” on A”. write R as a subset of $A \times A$. Also, find the domain and range of R. (Sep. 2021)
 - Let $A = \{1,2,3\}$ and $B = \{x|x \text{ is a prime number less than } 10\}$. Find $A \times B$ and $B \times A$ (May 2022)
 - The arrow diagram shows in figure a relationship between the sets P and Q. Write the relation in
(i) Set builder form (ii) Roster form
(iii) What is the domain and range of R. (May 2022)
- 
- If $A = \{5,6\}$, $B = \{4,5,6\}$, $C = \{5,6,7\}$. Show that $A \times A = (B \times B) \cap (C \times C)$. (Aug. 2022)
 - If $B \times A = \{(-2,3), (-2,4), (0,3), (0,4), (3,3), (3,4)\}$ find A and B (April 2023)
 - Find 'k' if $\text{fof}(k) = 5$ where $f(k) = 2k - 1$. (April 2023)
 - A Relation R is given by the set $\{(x, y)/y = x + 3, x \in \{0,1,2,3,4,5\}$ Determine its domain and the range. (April 2026) (June 2023)
 - Check whether $(f \circ g) = (g \circ f)$ if $f(x) = x - 6$ and $g(x) = x^2$ (June 2023)
 - If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B. (April 2024)
 - If $f(x) = 3x - 2$, $g(x) = 2x + k$ and $f \circ g = g \circ f$, then find value of k. (April 2024)
 - If $B \times A = \{(-2,3), (-2,4), (0,3), (0,4), (3,3), (3,4)\}$ find A and B. (July 2024)
 - Given $f(x) = 2x - x^2$. Find (i) $f(1)$ (ii) $f(x + 1)$. (July 2024)
 - Let $A = \{1,2,3\}$ and $B = \{x | x \text{ is a prime number less than } 10\}$. Find $A \times B$. (April 2025)
 - Find $f \circ g$ and $g \circ f$ when $f(x) = 2x + 1$ and $g(x) = x^2 - 2$. (April 2025)
 - If $A = \{2, -2,3\}$, $B = \{1, -4\}$, then find $A \times B, B \times A$. (July 2025)
 - The arrow diagram shows the relationship between the sets P and Q. Write the relation in
(i) Set builder form (ii) Roster form
 - Let $X = \{-5,1,3,4\}$ and $Y = \{a,b,c\}$. Determine whether the relation $R = \{(-5, a)(1, a)(3, b)\}$ is a function from X to Y? (April 2026)

**CHAPTER -2 .NUMBERS AND SEQUENCES**

- If m, n , are natural numbers, for what values of m , does $2^n \times 5^m$ ends in 5? (Sep. 2019)
- Find the 3^{rd} and 4^{th} terms of a sequence, if $a_n = \begin{cases} n^2 & \text{if } n \text{ is odd} \\ \frac{n^2}{2} & \text{if } n \text{ is even} \end{cases}$ (Sep 2019)
- Find the value of $1^2 + 2^2 + 3^2 + \dots + 10^2$ and hence deduce $2^2 + 4^2 + 6^2 + \dots + 20^2$ (Sep. 2019)
- Find the number of terms in the A.P. 3, 6, 9, 12, ... 111. (Sep. 2021)
- If $3 + k, 18 - k, 5k + 1$ are in A.P. then find the value of 'k'. (Sep. 2021)

6. Find the least number that is divisible by the first ten natural numbers. (Aug. 2022 & June 2023)
7. Find the 19^{th} term of an A.P. $-11, -15, -19, \dots$ (Aug. 2022)
8. If $13824 = 2^a \times 3^b$ then find 'a' and 'b' (May 2022)
9. Which term of an A.P. $16, 11, 6, 1, \dots$ is -54 ? (July 2024) (May 2022)
10. Find 'x' so that $x + 6, x + 12$ and $x + 15$ are consecutive terms of a Geometric Progression. (April 2023)
11. If $p^2 \times q^1 \times r^4 \times s^3 = 3,15,000$ then find the values of 'p', 'q', 'r' and 's' (April 2023)
12. Find the 8^{th} term of the G.P. $9, 3, 1, \dots$ (April 2026) (July 2024) (June 2023)
13. 'a' and 'b' are two positive integers such that $a^b \times b^a = 800$. Find 'a' and 'b' (July 2025) (April 2024)
14. Find the HCF of 23 and 12. (April 2024)
15. A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over. (April 2025)
16. Check whether the sequence $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \sqrt{50}, \dots$ is an A.P or not?. If it is an A. P, then find the common difference. (April 2025)
17. In a G.P. $729, 243, 81 \dots$ find t_7 (July 2025)
18. Find the least positive value of x such that $71 \equiv x \pmod{8}$ (April 2026)

CHAPTER -3 .ALGEBRA

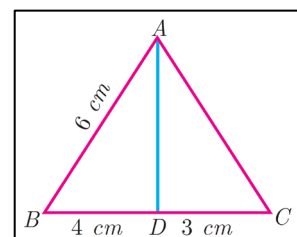
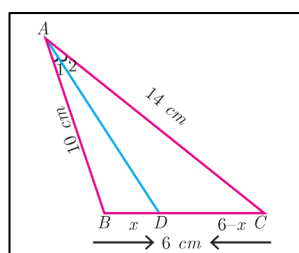
1. Find the value of k for which the equation $9x^2 + 3kx + 4 = 0$ has real and equal roots. (Sep. 2019)
2. If $A = \begin{pmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{pmatrix}$ then find the transpose of $-A$. (Sep. 2019)
3. Determine the quadratic equation whose sum and product of roots are -9 and 20. (Sep. 2021)
4. Determine the nature of the roots for the quadratic equation $15x^2 + 11x + 2 = 0$ (Sep. 2021)
5. Find the value of 'x', in $x^2 - 4x - 12 = 0$ (Aug. 2022)
6. Find the square root of the following rational expressions. $\frac{400x^4y^{12}z^{16}}{100x^8y^4z^4}$ (Aug. 2022)
7. If $P = \frac{x}{x+y}, Q = \frac{y}{x+y}$ then find $\frac{1}{(P^2 - Q^2)}$ (May 2022)
8. Find the excluded values of the following expression $\frac{7p+2}{8p^2+13p+5}$ (May 2022)
9. Simplify: $\frac{x+2}{4y} \div \frac{x^2-x-6}{12y^2}$. (April 2023)
10. Determine the nature of roots for the following quadratic equation. $2x^2 - x - 1 = 0$ (April 2023)
11. Determine the nature of the roots for the quadratic equation $15x^2 + 11x + 2 = 0$. (June 2023)
12. If $A = \begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$ then verify $(A^T)^T = A$ (June 2023)
13. Simplify- $\frac{4x^2y}{2z^2} \times \frac{6xz^3}{20y^4}$. (April 2024)
14. Find the sum and product of the roots for following quadratic equation $x^2 + 8x - 65 = 0$. (April 2024)
15. Find the LCM of $9a^3b^2, 12a^2b^2c$. (July 2024)

16. Determine the nature of the roots for the following quadratic equation $15x^2 + 11x + 2 = 0$. (July 2024)
19. Add: $\frac{x^3}{x-y} + \frac{y^3}{y-x}$. (April 2025)
20. If $A = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ find AB and BA . Verify if $AB = BA$. (April 2025)
21. Find the excluded values of the expression $\frac{t}{t^2-5t+6}$, if any. (July 2025)
22. Given that $A = \begin{pmatrix} -1 & 1 \\ a & b \end{pmatrix}$ where $a, b \geq 0$ find the values of a and b if $A^2 = I$. (July 2025)
23. Reduce the rational expression $\frac{x^2-16}{x^2+8x+16}$ to its lowest form. (April 2026)
24. If the difference between a number and its reciprocal is $\frac{24}{5}$ find the number. (April 2026)
25. Construct a 3×3 matrix whose elements are given by $a_{ij} = |i - 2j|$. (April 2026)

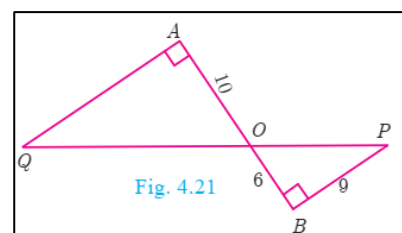
CHAPTER -4 .GEOMETRY

1. Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following $AB = 5\text{ cm}$, $AC = 10\text{ cm}$
 $BD = 1.5\text{ cm}$ and $CD = 3.5\text{ cm}$ (Sep. 2019)
2. In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$ If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15\text{ cm}$ find AE . (Sep. 2021)
3. $ABCD$ is a trapezium in which $AB \parallel DC$ and P, Q are points on AD and BC respectively, such that $PQ \parallel DC$ if $PD = 18\text{ cm}$, $BQ = 35\text{ cm}$ and $QC = 15\text{ cm}$, find AD . (Aug. 2022)
4. In the figure AD is the bisector of $\angle A$. If $BD = 4\text{ cm}$, $DC = 3\text{ cm}$ and $AB = 6\text{ cm}$, find AC . (April 2025) (May 2022)

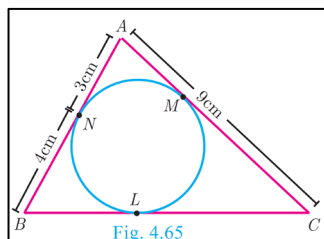
5. In the figure AD is the bisector of $\angle BAC$, if $AB = 10\text{ cm}$, $AC = 14\text{ cm}$ and $BC = 6\text{ cm}$. Find BD and DC . (April. 2023)



6. Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following $AB = 4\text{ cm}$, $AC = 6\text{ cm}$
 $BD = 1.6\text{ cm}$ and $CD = 2.4\text{ cm}$. (June 2023)
7. A man goes to 18 m due east and then 24 m due North. Find the distance of his current position from the starting point. (April 2024)
8. In figure QA and PB are perpendicular to AB . If $AO = 10\text{ cm}$,
 $BO = 6\text{ cm}$ and $PB = 9\text{ cm}$. Find AQ . (July 2024)



9. In the given figure, $\triangle ABC$ is circumscribing a circle. Find the length of BC . (July 2025)



10. D and E are respectively the points on the sides AB and AC of a $\triangle ABC$ such that $AB = 5.6$ cm, $AD = 1.4$ cm, $AC = 7.2$ cm and $AE = 1.8$ cm, show that $DE \parallel BC$. **(April 2026)**

CHAPTER -5.CO-ORDINATE GEOMETRY

1. Find the slope of a line joining the points $(14, 10)$ and $(14, -6)$. **(April 2026)** **(Sep. 2019)**
2. Show that the given points are collinear $(-3, -4)$, $(7, 2)$ and $(12, 5)$ **(Sep. 2021)**
3. Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$ **(Sep. 2021)**
4. Find the intercepts made by the following lines on the coordinate axes. $3x - 2y - 6 = 0$ **(Sep. 2021)**
5. If area of triangle formed by vertices $A(-1, 2)$, $B(k, -2)$ and $C(7, 4)$ is 22 sq. units.
find the value of ' k ' **(Aug. 2022)**
6. The line p passes through the points $(3, -2)$, $(12, 4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q ? **(Aug. 2022)**
7. Find the slope of a line joining the given points $(5, \sqrt{5})$ with the origin **(Aug. 2022)**
8. Show that the points $P(-1.5, 3)$, $Q(6, -2)$, $R(-3, 4)$ are collinear. **(May 2022)**
9. The line p passes through the points $(3, -2)$, $(12, 4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q ? **(May 2022)**
10. Find the equation of a straight line which has slope $-\frac{5}{4}$ and passing through the point $(-1, 2)$ **(May 2022)**
11. A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at $(5, 11)$. The cat wishes to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk. **(April 2023)**
12. If the straight lines $12y = -(p + 3)x + 12$, $12x - 7y = 16$ are perpendicular then find ' p ' **(April 2023)**
13. Find the equation of a straight line which is parallel to the line $3x - 7y = 12$ and passing through the point $(6, 4)$. **(June 2023)**
14. Find the slope of a line joining the given points $(5, \sqrt{5})$ with the origin. **(June 2023)**
15. If the points $A(-3, 9)$, $B(a, b)$ and $C(4, -5)$ are collinear and if $a + b = 1$, then find a and b . **(April 2024)**
16. Find the equation of a straight line which has slope $-\frac{5}{4}$ and passing through the points $(-1, 2)$. **(April 2024)**
17. The line p passes through the points $(3, -2)$, $(12, 4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q ? **(July 2024)**
18. Find the slope of the straight line $6x + 8y + 7 = 0$. **(July 2024)**

26. The line through the points $(-2, a)$ and $(9, 3)$ has slope $-\frac{1}{2}$. Find the value of 'a'. (April 2025)
27. Show that the straight lines $x - 2y + 3 = 0$ and $6x + 3y + 8 = 0$ are perpendicular. (April 2025)
28. Find the slope of a line joining the points $(-6, 1)$ and $(-3, 2)$. (July 2025)
29. Find the intercepts made by the line $4x + 3y + 12 = 0$ on the coordinate axes. (July 2025)
30. Find the equation of a straight line whose intercepts on the x and y axes are 4 and -6 respectively. (April 2026)

CHAPTER -6 .TRIGONOMETRY

1. Prove that $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$ (Sep. 2019)
2. Find the angle of elevation of the top of a tower from a point on the ground, which is $30m$ away from the foot of a tower of height $10\sqrt{3}m$. (Sep. 2021 & Aug. 2022)
3. From the top of a rock $50\sqrt{3}m$ height, the angle of depression of a car on the ground is observed to be 30° . Find the distance of the car from the rock. (April 2025) (May.2022)
4. Prove that $\frac{\sec\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} = \cot\theta$ (April 2023)
5. Prove that $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$ (June 2023)
6. Prove that $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta + \cot\theta$. (April 2026) (April 2024)
7. Prove the following identity $\frac{1-\tan^2\theta}{\cot^2\theta-1} = \tan^2\theta$. (July 2024)
8. Prove $\frac{\cos\theta}{1+\sin\theta} = \sec\theta - \tan\theta$ (July 2025)

CHAPTER -7 .MESURATION

1. Find the volume of the iron used to make a hollow cylinder of height 9 cm and whose internal and external radii are 3 cm and 5 cm respectively. (Sep. 2019)
2. Find the diameter of a sphere whose surface area is 154 m^2 (Sep. 2019)
3. If the base area of a hemispherical solid is 1386 sq. meters , then find its total surface area? (Sep. 2019)
4. The heights of two right circular cones are in the ratio $1: 2$ and the perimeters of their bases are in the ratio $3: 4$. Find the ratio of their volumes. (Sep. 2021)
5. Find the volume of a cylinder whose height is $2m$ and whose base area is 250 m^2 (Sep. 2021)
6. Find its radius and height. radius 7 cm is 704 cm^2 , then find its slant height. (Aug. 2022)
7. If the total surface area of a cone of The radius and height of a cylinder are in the ratio $5: 7$ and its curved surface area is 5500 sq. cm . (Aug. 2022)
8. The radius of a spherical balloon increases from 12 cm to 16 cm as air being pumped into it. Find the ratio of the surface area of the balloons in the two cases. (May 2022)
9. The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of heights. (May 2022)

10. The radius of a conical tent is $7m$ and height is $24m$. Calculate the length of the canvas used to make the tent if the width of the rectangular canvas is $4m$? (April 2023)
11. If the ratio of radii of two spheres is $4:7$, find the ratio of their volumes. (April 2025) (April 2023)
12. The curved surface area of a right circular cylinder of height $14cm$ is $88cm^2$ Find the diameter of the cylinder. (July 2025) (June 2023)
13. The volume of a solid right circular cone is $11088 cm^3$ If its height is $24cm$ then find the radius of the cone. (June 2023)
14. If the base area of a hemispherical solid is $1386 sq. meters$, then find its total surface area? (April 2024)
15. Find the volume of a cylinder whose height is $2m$ and whose base area is $250 m^2$ (April 2024)
16. A cylindrical drum has a height of $20 cm$ and base radius of $14 cm$. Find its curved surface Area?. (July 2024)
17. The volumes of two cones of same base radius are $3600 cm^3$ and $5040 cm^3$. Find the ratio of heights. (July 2024)
18. The slant height of a frustum of a cone is $5 cm$ and the radii of its ends are $4 cm$ and $1 cm$. Find its curved surface area. (July 2025) (April 2025)
19. Find the ratio of the volume of a cube to that of a largest sphere which exactly fits into the cube. (April 2026)

CHAPTER -8 .STATISTICS AND PROBABILITY

1. Find the range and coefficient of range of the following data $63, 89, 98, 125, 79, 108, 117, 68$. (Sep. 2019)
2. A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head. (Sep. 2021)
3. A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is (i) white ball (ii) black or red ball (Aug. 2022)
4. Two coins are tossed together. What is the probability of getting different faces of the coins? (May. 2022)
5. Find the range and coefficient of range of the following data $63, 89, 98, 125, 79, 108, 117, 68$. (April 2023)
6. A and B are two candidates seeking admission to IIT. The probability that A getting selected is 0.5 and the probability that both A and B getting selected is 0.3 . Prove that the probability of B being selected is at most 0.8 . (April 2023)
7. Find the standard deviation of first 21 natural numbers. (April 2026) (June 2023)
8. A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head. (April 2026) (June 2023)
9. Find the Range and Co-efficient of range of the following data : $25, 67, 48, 53, 18, 39, 44$. (April 2024)
10. What is the Probability that a leap year selected at random will contain 53 Saturdays?. (April 2024)
11. A Coin is tossed thrice. What is the Probability of getting two Consecutive tails?. (July 2024)

12. Find the range of the following distribution :

(April 2025)

Age (in Years)	16 – 18	16 – 18	16 – 18	16 – 18	16 – 18	16 – 18
Number of Students	0	4	6	8	2	2

13. A flower is selected at random from a basket containing 80 yellow, 70 red and 50 white flowers. Find the probability of selecting a yellow coloured flower.

(April 2025)

14. If the range and the smallest value of a set of data are 36.8 and 13.4 respectively, then find The largest value.

(July 2025)

15. Two coins are tossed together. What is the probability of getting different faces on the coins?

(July 2025)

10TH MATHS PUBLIC EXAM QUESTIONS COLLECTIONS

5 MARK QUESTIONS EM NEW (2026 - 2027)

CHAPTER -1.RELATIONS AND FUNCTIONS

1. Let A =The set of all natural numbers less than 8, B =The set of all prime numbers less than 8,

C =The set of even prime number. Verify $(A \cap B) \times C = (A \times C) \cap (B \times C)$ (July 2024) (Sep. 2019)

2. Let $A = \{x \in W | x < 2\}$, $B = \{x \in N | 1 < x \leq 4\}$ and $C = \{3,5\}$. Verify that

$A \times (B \cap C) = (A \times B) \cap (A \times C)$ (April 2026) (Sep. 2021)

3. Let A =The set of all natural numbers less than 8, B =The set of all prime numbers less than 8, C = The set of even prime number. Verify $A \times (B - C) = (A \times B) - (A \times C)$

(May 2022)

4. Let $A = \{1,2,3,4\}$ and $B = \{2,5,8,11,14\}$ be two sets. Let $f: A \rightarrow B$ be a function give by $f(x) = 3x - 1$.

Represent this function

(July 2025) (Sep. 2019)

(i) by arrow diagram (ii) in a table form

(iii) as a set of ordered pairs (iv) in a graphical form

5. Represent each of the given relations by (a) an arrow diagram, (b) a graph and (c) a set in roster from,

wherever possible. $\{(x, y) | y = x + 3, x, y \text{ are natural numbers} < 10\}$

(Aug. 2022)

6. Let $f: A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, wehre $A = \{2,4,6,10,12\}$, $B = \{0,1,2,4,5,9\}$.

Represent f by

(April 2025) (April 2023)

(i) Set of ordered pairs, (ii) a table (iii) a graph (iv) an arrow diagram

7. Let $A = \{x \in W | x < 3\}$, $B = \{x \in N | 1 < x \leq 5\}$ and $C = \{3,5,7\}$.

Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(April 2023)

8. Let $A = \{x \in W | x < 2\}$, $B = \{x \in N | 1 < x \leq 4\}$ and $C = \{3,5\}$.

Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(Sep. 2023)

9. Let $A = \{x \in N | 1 < x < 4\}$, $B = \{x \in W | 0 \leq x < 2\}$ and $C = \{x \in W | x < 3\}$.

(April 2024)

Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$

10. Let $A = \{0,1,2,3\}$ and $B = \{1,3,5,7,9\}$ be two sets. Let $f: A \rightarrow B$ be a function give by $f(x) = 2x - 1$.

Represent this function

(April (2024) (Sep. 2019)

(i) by arrow diagram (ii) in a table form

(iii) as a set of ordered pairs (iv) in a graphical form

11. A function $f: [-5, 9] \rightarrow \mathbb{R}$ is defined as follows:

(July 2024)

$$f(x) = \begin{cases} 6x + 1; & -5 \leq x < 2 \\ 5x^2 - 1; & 2 \leq x < 6 \\ 3x - 4; & 6 \leq x \leq 9 \end{cases}$$

Find (i) $f(-3) + f(2)$

(ii) $f(7) - f(1)$

(iii) $2f(4) + f(8)$

(iv) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$

12. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8,

C = The set of even prime numbers. Verify that $(A \cap B) \times C = (A \times C) \cap (B \times C)$. (April (2025))

13. If $f(x) = x - 1$, $g(x) = 3x + 1$ and $h(x) = x^2$, then show that

$$(f \circ g) \circ h = f \circ (g \circ h)$$

(April 2026) (July 2025)

CHAPTER -2 .NUMBERS AND SEQUENCES

1. Find the sum of all natural numbers between 100 and 1000 which are divisible by 11. (Sep. 2019)

2. The sum of three consecutive terms that are in A.P. is 27 and their product is 288.

Find the three terms. (Sep. 2021)

3. Find the HCF of 396, 504, 636 (July 2025) (Sep. 2021)

4. Find the largest number which divides 1230 and 1926 leaving remainder 12 in each case. (Aug. 2022)

5. In an A.P nine times ninth term is equal to the fifteen times fifteenth term, show that six times twenty fourth term is zero. (Aug. 2022)

6. If l^{th} , m^{th} and n^{th} terms of an A.P. are, x , y , z respectively, then show that (May 2022)

$$(i) x(m - n) + y(n - l) + z(l - m) = 0 \quad (ii) (x - y)n + (y - z)l + (z - x)m = 0$$

7. The ratio of 6^{th} and 8^{th} term of an A.P. is 7: 9. Find the ratio 9^{th} term to 13^{th} term. (May 2022)

8. The houses of a street are numbered from 1 to 49. Senthil's house is numbered Such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number?. (April 2023)

9. Find the sum to n terms of the series $5 + 55 + 555 + \dots$ (April 2023)

10. Find the sum to n terms of the series $3 + 33 + 333 + \dots$ (April 2026) (July 2024)(June 2023)

11. Rekha has 15 square Colour papers of sizes 10 cm, 11 cm, 12 cm, ... 24 cm. How much area can be decorated with these Colour papers. (July 2025) (July 2024)(June 2023)

12. Find the Sum of $9^3 + 10^3 + \dots + 21^3$. (April 2024)

13. Find the sum to n terms of the series $7 + 77 + 777 + \dots$ (April 2024)

14. If $p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 113400$ where p_1, p_2, p_3, p_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers, find the value of p_1, p_2, p_3, p_4 and x_1, x_2, x_3, x_4 . (April 2025)

15. Find the sum of the series $6^2 + 7^2 + 8^2 + \dots + 21^2$. (April 2025)

16. Find the sum of all natural numbers between 300 and 600 which are divisible by 7. (April 2026)

CHAPTER -3 .ALGEBRA

1. Solve: $6x + 2y - 5z = 13$, $3x + 3y - 2z = 13$, $7x + 5y - 3z = 26$ (Sep. 2019)

2. Find the GCD of the given polynomials $x^4 + 3x^3 - x - 3$ and $x^3 + x^2 - 5x + 3$ (Sep. 2019)

3. Find the square root of the following polynomials by division method. $\frac{x^2}{y^2} - \frac{10x}{y} + 27 - \frac{10y}{x} + \frac{y^2}{x^2}$
4. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$, and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$ (Sep. 2019)
5. If the equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots, then prove that $c^2 = a^2(1 + m^2)$ (Sep. 2021)
6. Solve $x + y + z = 5, 2x - y + z = 9, x - 2y + 3z = 16$ (Sep. 2021)
7. Find the square root of $64x^2 - 16x^3 + 17x^2 - 2x + 1 = 0$ (Sep. 2021)
8. Simplify: $\frac{b^2+3b-28}{b^2+4b+4} \div \frac{b^2-49}{b^2-5b-14}$ (Aug. 2022)
9. Find the square root of $x^4 - 12x^3 + 42x^2 - 36x + 9$ (Aug. 2022)
10. Solve $x^2 + 2x - 2 = 0$ by Formula method. (Aug 2022)
11. If $36x^4 - 60x^3 + 61x^2 - mx + n$ is a perfect square, find the values of 'm' and 'n' (May 2022)
12. Solve $pqx^2 - (p + q)^2x + (p + q)^2 = 0$ (May 2022)
13. If α and β are the roots of $7x^2 + ax + 2 = 0$ and $\beta - \alpha = \frac{-13}{7}$. Find the values of a . (May 2022)
14. Solve: $x + 20 = \frac{3y}{2} + 10 = 2z + 5 = 110 - (y + z)$ (April 2023)
15. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, and $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$ (July 2024) (April 2023, 2026)
16. Solve $3x - 2y + z = 2, 2x + 3y - z = 5, x + y + z = 6$. (June 2023)
17. Find the square root of the following polynomials by division method.
 $121x^4 - 198x^3 - 183x^2 + 216x + 144$ (June 2023)
18. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$ (July 2025) (April 2024) (June 2023)
19. Find the square root of $64x^2 - 16x^3 + 17x^2 - 2x + 1 = 0$. (April 2024)
20. If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the values of 'a' and 'b' (July 2024)
31. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$, show that $(AB)^T = B^T A^T$. (April 2025)
32. Determine the nature of the roots of the equation $(x - 1)^2 + (x - 2)^2 + (x - 3)^2 = 0$ (April 2025)
33. Find the values of m and n of the polynomial $36x^4 - 60x^3 + 61x^2 - mx + n$ are perfect squares. (July 2025)
34. If α and β are the roots of the equation $2x^2 - 3x + 1 = 0$, then form a quadratic equation whose roots are $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$. (April 2026)

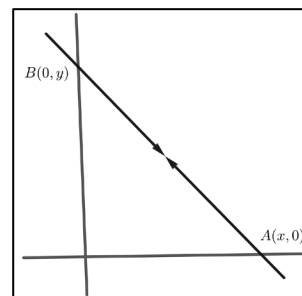
CHAPTER -4 .GEOMETRY

1. State and prove Angle Bisector Theorem. April (2025) (Sep. 2019, Aug. 2022 April 2023)
2. State and prove Pythagoras Theorem. (July 2025) (July 2024)(Sep. 2021) (June 2023)
3. Show that in a triangle, the medians are concurrent. (Sep. 2021)

4. A man goes 18 m due east and then 24 m due north. Find the distance of his current position from the starting point?. (Aug. 2022)
5. State and Prove Basic Proportionality Theorem (BPT) or Thales theorem. (May 2022)
6. An Aeroplane leaves an airport and flies due north at a speed of 1000 km/hr . At the same time, another Aeroplane leaves the same airport and flies due west at the speed of 1200 km/hr . How far apart will be the two planes after $1\frac{1}{2}$ hours?. (May 2022)
7. Two poles of height ' a ' meters and ' b ' meters are ' p ' meters apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ meters. (April 2023)
8. State and Prove Thales Theorem. (April 2024)
9. Show that the angle bisectors of a triangle are concurrent. (April 2026)

CHAPTER -5.CO-ORDINATE GEOMETRY

1. A straight line AB cuts the co-ordinate axes at A and B. If the mid-point of AB is (2, 3) , find the equation of AB. (Sep. 2019)
2. Find the value of ' k ' if the area of quadrilateral is 28 sq. units, whose vertices are $(-4, -2)$, $(-3, k)$, $(3, -2)$ and $(2, 3)$ (Sep. 2019)
3. Find the equation of the median and altitude of $\triangle ABC$ through A where the vertices are $A(6,2)$, $B(-5, -1)$ and $C(1,9)$ (Sep. 2021)
4. If the points $P(-1, -4)$, $Q(b, c)$ and $R(5, -1)$ are collinear and if $2b + c = 4$, then find the values of ' b ' and ' c ' (Sep. 2021)
5. A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at $(5, 11)$. The cat wishes to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk. (Aug 2022)
6. Find the area of the quadrilateral formed by the points $(8, 6)$, $(5, 11)$, $(-5,12)$ and $(-4,3)$. (April 2026) (Aug. 2022)
7. Find the equation of the line passing through $(22, -6)$ and having intercept on X - axis exceeds the intercept on Y -axis by 5 units. (May 2022)
8. A quadrilateral has vertices $A(-4, -2)$, $B(5, -1)$, $C(6,5)$ and $D(-7,6)$. Show that the mid-points of its sides form a parallelogram. (May 2022)
9. Find the area of the quadrilateral formed by the points $(8, 6)$, $(5, 11)$, $(-5,12)$ and $(-4,3)$ (Apr. 2023)



10. Find the equation of a straight line parallel to X —axis and passing through the point of intersection of the lines $7x - 3y = -12$ and $2y = x + 3$ **(April 2023)**
12. Find the area of the quadrilateral whose vertices are at $(-9, -2)$, $(-8, -4)$, $(2, 2)$ and $(1, -3)$ **(July 2025) (April 2024) (June 2023)**
13. Find the equation of the perpendicular bisector of the line joining the points $A(-4, 2)$ and $B(6, -4)$. **(April 2024) (June 2023)**
14. Find the area of the quadrilateral whose vertices are at $(-9, 0)$, $(-8, 6)$, $(-1, -2)$ and $(-6, -3)$. **(July 2024)**
15. Find the equation of a straight line through the intersection of lines $7x + 3y = 10$, $5x - 4y = 1$ and parallel to the line $13x + 5y + 12 = 0$. **(July 2024)**
17. Find the value of k , if the area of a quadrilateral is 28 sq. units, whose vertices are taken in the order $(-4, -2)$, $(-3, k)$, $(3, -2)$ and $(2, 3)$. **(April 2025)**
18. A line makes positive intercepts on coordinate axes whose sum is 7 and it passes through $(-3, 8)$. Find its equation. **(April 2025)**
19. Find the equation of a straight line perpendicular to $2x - 3y + 6 = 0$ and passes through the midpoint of the line segment joining the points $(-4, -7)$ and $(6, 5)$. **(July 2025)**
20. $A(-3, 0)$, $B(10, -2)$ and $C(12, 3)$ are the vertices of $\triangle ABC$. Find the equation of the altitude through A. **(April 2026)**

CHAPTER -6 .TRIGONOMETRY

1. From the top of the tower 60m high the angles of depression of the top and bottom of a vertical lamp post are observed to be 38° and 60° respectively. Find the height of the lamp post. ($\tan 38^\circ = 0.7813$, $\sqrt{3} = 1.732$) **(Sep. 2019)**
2. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$) **(April 2026) (Sep. 2021)**
3. To a man standing outside his house, the angles of elevation of the top and bottom of a window are 60° and 45° respectively. If the height of the man is 180 cm and if he is 5m away from the wall, what is the height of the window ?.. ($\sqrt{3} = 1.732$) **(Aug. 2022)**
4. From a point on the ground, the angles of elevation of the bottom and top of a tower fixed at the top of a 30 m high building are 45° and 60° respectively. Find the height of the tower. ($\sqrt{3} = 1.732$) **(May 2022)**
10. From the top of a lighthouse, the angle of depression of two ships on the opposite sides of it are observed to be 30° and 60° . If the height of the lighthouse is h meters and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\frac{4h}{\sqrt{3}}m$. **(April 2023)**
5. Prove that $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = 2 \sec \theta$ **(June 2023)**

6. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$) (April 2024) (June 2023)
7. A pole 5 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point 'A' on the ground is 60° and the angle of depression to the point 'A' from the top of the tower is 45° . Find the height of the tower. ($\sqrt{3} = 1.732$) (July 2024)
8. Prove that: $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$. (April 2025)
9. From the top of a 12 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30° . Determine the height of the tower. (July 2025)

CHAPTER -7 .MESURATION

1. A cylindrical glass with diameter 20 cm has water to a height of 9cm. A small cylindrical metal of radius 5cm and height 4cm is immersed it completely. Calculate the rise of the water in the glass. (Sep. 2019)
2. If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7cm, find the volume of the frustum. (April 2024) (Sep. 2021)
3. A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25cm. Find the total surface area of the toy if its common diameter is 12cm. (July 2025) (Sep. 2021)
4. A cylindrical drum has height of 20cm and base radius of 14cm. Find its curved surface area and the total surface area. (Aug. 2022)
5. If the circumference of base of a conical wooden piece is 484 cm then find its volume when its height is 105cm. (Aug. 2022)
6. A container open at the top is in the form of a frustum of a cone of height 16cm with radii of its lower and upper ends are 8cm and 20cm respectively. Find the cost of milk which can completely fill a container at the rate of 540 per liter. (May 2022)
7. Nathan, an engineering student was asked to make a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3cm and its length 12cm. If each cone has height 2cm, find the volume of the model that Nathan made. (April 2025) (May 2022)
8. The radius and height of a cylinder are in the ratio 5: 7 and its curved surface area is 5500 sq. cm. Find its radius and height. (April 2023)
9. Arul has to make arrangements for the accommodation of 150 persons for his family function. For this purpose, he plans to build a tent which is in the shape of cylinder surmounted by cone. Each person occupies 4 sq. m of the space on ground and 40 cu. Meter of air to breathe. What should be the height of the conical part of the tent if the height of cylindrical part is 8m? (April 2023)
10. A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2cm. How many small spheres can be obtained ? (April 2025) (July 2024) (June 2023)

11. A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice-cream. The ice-cream is to be filled in cones of height 9 cm and base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container. **(April 2024)**
12. The internal and external diameters of a hollow hemispherical vessel are 20 cm and 28 cm respectively. Find the cost to paint the vessel all over at ₹0.14 per . cm² . **(July 2024)**
13. An Aluminum sphere of radius 12 cm is melted to make a cylinder of radius 8 cm. Find the height of the cylinder. **(July 2025)**
14. A garden roller whose length is 3m long and whose diameter is 2.8m is rolled to level a garden. How much area will it cover in 8 revolutions? **(April 2026)**

CHAPTER -8 .STATISTICS AND PROBABILITY

1. The scores of a cricketer in 7 matches are 70, 80, 60, 60, 40, 90, 95. Find the standard deviation **(Sep. 2019)**
2. Two unbiased dice are rolled once. Find the probability of getting **(Sep. 2019, Aug. 2022 & Apr. 2023)**
 - (i) a doublet (equal numbers on both dice)
 - (ii) the product as a prime number
 - (iii) the sum as a prime number
 - (iv) the sum as 1
3. Two dice are rolled. Find the probability that sum of outcomes is
 - (i) equal to 4,
 - (ii) greater than 10,
 - (iii) less than 13**(Sep. 2021)**
4. In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that
 - (i) The student opted for NCC but not NSS.
 - (ii) The student opted for NSS but not NCC.
 - (iii) The student opted for exactly one of them.**(May 2022)**
5. Two dice are rolled once. Find the probability of getting an even number on the first die or a total of face sum 8. **(April 2024) (June 2023)**
6. A teacher asked the students to complete 60 pages of a record note book. Eight students have completed only 32, 35, 37, 30, 33, 36, 35 and 37 pages. Find the standard deviation of the pages completed by them. **(July 2024)**
7. Two dice are rolled. Find the probability that the sum of outcomes is (i) equal to 4 (ii) greater than 10 (iii) less than 13. **(July 2024)**
8. Find the coefficient of variation 24, 26, 33, 37, 29, 31. **(April 2026) (April 2024)**
9. The total marks scored by two students, Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance. **(April 2025)**
10. Two dice are rolled together. Find the Probability of getting a doublet or sum of faces as 4. **(April 2026)**
11. A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is random from the bag. Find the probability that the ball drawn is
 - (i) white
 - (ii) black or red
 - (iii) not white
 - (iv) neither white nor black**(April 2025)**

12. The marks scored by the students in a slip test are given below. Find the standard deviation of their marks. **(July 2025)**

x	4	6	8	10
f	7	3	5	9

10TH MATHS PUBLIC GEOMETRY QUESTIONS COLLECTIONS

8 MARK QUESTIONS EM NEW (2026 - 2027)

- Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC (scale factor $\frac{6}{5} > 1$) **(Sep. 2019)**
- Draw the two tangents from a point which is 10cm away from the centre of a circle of radius 5cm. Also, measure the lengths of the tangents. **(July 2024)(Sep. 2019 & June 2023)**
- Construct a ΔPQR which the base $PQ = 4.5cm$, $\angle R = 35^\circ$ and the median from R to PQ is 6cm. **(Sep. 2021) (April 2024)**
- Draw a circle of diameter 6cm from a point P , which 8cm away from its centre. Draw two tangents PA and PB to the circle and measure their lengths. **(April 2026) (April 2024) (Sep. 2021 & Aug. 2022)**
- Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle PQR (scale factor $\frac{7}{3} > 1$) **(Aug. 2022)**
- Construct a ΔABC such that $AB = 5.5cm$, $\angle C = 25^\circ$ and the altitude from C to AB is 4cm. **(May 2022)**
- Draw two tangents from a point which is 5cm away from the center of a circle of diameter 6cm. Also, measure the lengths of the tangents. **(July 202) (May 2022)**
- Take a point which is 11 cm away from the center of a circle of radius 4cm and draw the two tangents to the circle from the point. **(April 2023)**
- Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR (scale factor $\frac{3}{5} < 1$) **(April 2026) (June 2023)**
- Draw a triangle ABC of base $BC = 8cm$, $\angle A = 60^\circ$ and the bisector of $\angle A$ meets BC at D such that $BD = 6cm$. **(April 2023)**
- Construct a ΔPQR in which $QR = 5$ cm, $\angle P = 40^\circ$ and the Median PG from P to QR is 4.4 cm. Find the length of the Altitude from P to QR . **(July 2024)**
- Draw a triangle ABC base $BC = 8$ cm, $\angle A = 60^\circ$ and the bisector of $\angle A$ meets BC at D such that $ED = 6$ cm. **(April 2025)**
- Take a point which is 11 cm away from the centre of a circle of radius 4 cm and draw the two tangents to the circle from that point **(April 2025)**

14. Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{4}$ of the corresponding sides of the triangle PQR. (scale factor $\frac{7}{4} > 1$) (July 2025)

10TH MATHS PUBLIC GRAPH QUESTIONS COLLECTIONS

8 MARK QUESTIONS EM NEW (2026 - 2027)

1. Graph the following quadratic equation $x^2 - 8x + 16 = 0$ and state their nature of solution. (Sep. 2019)
2. Draw the graph of $y = 2x^2 - 3x - 5$ and hence use it to solve $2x^2 - 4x - 6 = 0$ (April 2024) (Sep. 2019)
3. Draw the graph of $x^2 + x - 12 = 0$ and state the nature of their solution. (July 2025) (Sep. 2021)
4. Draw the graph of $y = x^2 + 3x - 4$ and hence use it to solve $x^2 + 3x - 4 = 0$ (Sep. 2021)
5. Graph the following quadratic equation $x^2 - 9x + 20 = 0$ and state their nature of solution. (Aug. 2022)
6. Draw the graph of $y = x^2 - 4x + 3$ and use it solve $x^2 - 6x + 9 = 0$ (Aug. 2022 & May 2022)
7. Draw the graph of $x^2 - 4x + 4 = 0$ and state the nature of their solution (May 2022)
9. Varshika drew 6 circles with different sizes. Draw a graph for the relationship between the diameter and circumference of each circle as shown in the table and use it to find the circumference of a circle when its diameter is 6 cm. (April 2023)

Diameter (x)cm	1	23	3	4	5
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5

10. Draw the graph of $y = x^2 - 5x - 6$ and hence use it to solve $x^2 - 5x - 14 = 0$ (April 2023)
11. Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also (i) find y when $x = 9$. (ii) find x when $y = 7.5$ (April 2026) (June 2023)
12. Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$ (June 2023)
13. Draw the graph of $xy = 24, x, y > 0$. Using the graph find, (i) y when $x = 3$ and (ii) x when $y = 6$. (April 2024)
14. Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$. (July 2024)
15. The following table shows the data about the number of pipes and the time taken to fill the same tank.

No. of. pipes x	2	3	6	9
Time Taken y (in mins)	45	30	15	10

Draw the graph for the above data and hence

(July 2024)

- (i) Find the time taken to fill the tank when five pipes are used.
- (ii) Find the number of pipes when the time is 9 minutes.
16. Graph the quadratic equation $x^2 - 9x + 20 = 0$ and state the nature of solution. (April 2025)

17. A bus is travelling at a uniform speed of 50 km/hr . Draw the distance - time graph and hence find.
 (i) the constant of variation. (April 2025)
 (ii) how far will it travel in 90 minutes?
 (iii) The time required to cover a distance of 300 km.
18. A school announces that for a certain competition the cash prize will be distributed for all the participants equally as shown below : (July 2025)

No .of Participants x	2	4	6	8	10
Amount for each participant y in ₹	180	90	60	45	36

- (i) Find the constant of variation.
 (ii) Graph the above data and hence find how much each participant will get if the number of participants are 12.
19. Graph the quadratic equation $x^2 - 6x + 9 = 0$ and state its nature of solutions. (April 2026)

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